

Adaptive Kernel Selection for Stein Variational Gradient Descent

Anonymous Authors

Abstract

A central challenge in Bayesian inference is efficiently approximating posterior distributions. Stein Variational Gradient Descent (SVGD) is a popular variational inference method which transports a set of particles to approximate a target distribution. The SVGD dynamics are governed by a reproducing kernel Hilbert space (RKHS) and are highly sensitive to the choice of the kernel function, which directly influences both convergence and approximation quality. The commonly used median heuristic offers a simple approach for setting kernel bandwidths, but its single scalar bandwidth is inflexible and, as our experiments make explicit, its accuracy degrades sharply as dimension grows: in our multivariate Gaussian benchmark the median heuristic recovers less than half of the leading-component variance (0.475 versus a target of 1.000 at $d = 8$), whereas the method we propose stays within a few percent of every target variance. In this work, we propose an alternative strategy for adaptively choosing kernel parameters over an abstract family of kernels. Recent convergence analyses based on the kernelized Stein discrepancy (KSD) suggest that optimizing the kernel parameters by maximizing the KSD can improve performance. Building on this insight, we introduce Adaptive SVGD (Ad-SVG), a method that alternates between updating the particles via SVGD and adaptively tuning kernel bandwidths through gradient ascent on the KSD. We provide a simplified theoretical analysis that extends existing results on minimizing the KSD for fixed kernels to our adaptive setting, showing convergence properties for the maximal KSD over our kernel class. Our empirical results reinforce this analysis with a clear and consistent margin: across a Gaussian mixture, an ODE-based inverse problem, Bayesian logistic regression, and Gaussian-process inference, Ad-SVG matches the median heuristic in the easy low-dimensional regime and decisively outperforms it wherever dimension or variance collapse make the problem hard, tracking the true posterior uncertainty where the standard heuristic systematically underestimates it.

1 Introduction

Stein Variational Gradient Descent (SVGD) [18] is a deterministic particle-based method for approximate Bayesian inference that has emerged as a popular alternative to traditional Markov Chain Monte Carlo (MCMC) methods. SVGD evolves a set of particles using update directions derived from the functional gradient of the Kullback-Leibler (KL) divergence to the target distribution, with updates constrained to lie within the unit ball of a reproducing kernel Hilbert space (RKHS). A critical limitation of SVGD is its sensitivity to kernel choice, which significantly influences the algorithm’s performance [8, 28]. The effect is not subtle: as we quantify below, sweeping the fixed bandwidth over several orders of magnitude changes the final Wasserstein-1 error by more than an order of magnitude on a simple one-dimensional target (Figure 1), and only a narrow band of bandwidths yields a usable approximation. Additionally, the resulting particle approximation commonly underestimates the posterior variance [2]. These observations have led to the widely

held belief that SVGD in general fails to perform well as the dimension of the underlying state space increases. In this work, we challenge this belief head-on: by introducing an adaptive mechanism that dynamically tunes the kernel during inference through maximization of the kernelized Stein discrepancy (KSD), we show that the same SVGD dynamics recover the posterior uncertainty accurately in settings where the standard heuristic collapses. The gains are consistent and grow with dimension—for example, our method keeps a χ^2 diagnostic close to its true expectation across $d = 2, \dots, 8$ while the median heuristic drifts progressively away—demonstrating that kernel choice alone, rather than any architectural change, accounts for much of SVGD’s reported difficulty in high dimensions.

1.1 Related work.

Since its introduction [18], SVGD has become a widely used tool for approximate Bayesian inference in a range of machine learning applications [22, 24, 30, 15]. Recent work has made substantial progress in understanding the theoretical underpinnings of SVGD. Mean-field convergence has been analyzed in both continuous-time [23, 8, 5] and discrete-time [16, 32] settings, while finite-particle convergence rates have been established under various assumptions [3, 33]. The SVGD dynamics have also been connected to gradient flows on probability distribution spaces [19, 8], drawing analogy to Jordan, Kinderlehrer, and Otto [14].

The performance of SVGD depends critically on the choice of the kernel function, as it determines the interaction between particles and the overall convergence of the method (see also Figures 1 and 9). Convergence results typically refer to mean-field convergence with respect to KSD, whose relation to weak convergence depends on the selected kernel [11]. The commonly used median heuristic [12] provides a simple implementation but lacks theoretical justification and exhibits a measurable, well-documented deterioration in approximation quality as the dimensionality of the problem increases [2, 40, 36]. Recent work has developed tools to mitigate performance degradation in high-dimensional settings [7, 37, 10, 21]. The algorithm has also been modified through the use of neural networks to learn the update direction [17, 39].

The approach most closely related to our work is Ai et al. [1], which introduces a mixture-of-kernels framework. Their method defines a KSD for a weighted linear combination of kernels and learns the kernel weights by maximizing this multiple-kernel KSD. However, their approach is confined to a finite kernel basis and to optimizing weights over that fixed basis, whereas we optimize continuous kernel parameters—most notably per-dimension bandwidths—directly, which is precisely the flexibility our high-dimensional comparisons show to be decisive.

1.2 Contributions.

In this work, we address the fundamental research question of whether SVGD, without any architectural or algorithmic enhancements, can achieve competitive performance when equipped with a principled and adaptively chosen kernel. By isolating kernel selection as the single variable that changes—holding the SVGD dynamics, step sizes, particle counts, and targets fixed—our experiments provide a clean, controlled measurement of how much of SVGD’s effectiveness is attributable to kernel choice alone, and how far adaptive kernel learning can push the capabilities of the original method.

Our main contributions are as follows.

- (i) **Adaptive Kernel Selection Method.** We propose a novel method that dynamically updates the kernel parameters by maximizing the KSD during SVGD inference. In contrast to the commonly used median heuristic, which relies on a single scalar bandwidth, our approach allows for the optimization of multiple continuous kernel parameters, enabling greater flexibility and adaptivity during SVGD updates.
- (ii) **Theoretical Analysis.** We provide theoretical motivation by analyzing our algorithm in the discrete-time mean-field setting, extending existing convergence results for SVGD with fixed kernels. Specifically, we show that the supremum of the KSD over a parameterized kernel class converges to zero as the particle distribution approaches the target. Assuming a Stein logarithmic Sobolev inequality we further derive iteration complexity in the mean-field limit.
- (iii) **Empirical Validation.** Across four benchmarks—a Gaussian mixture, an ODE-based inverse problem, Bayesian logistic regression, and Gaussian-process inference—our adaptive kernel selection matches the median heuristic on easy low-dimensional problems and outperforms it by a widening margin as dimension grows, directly correcting the variance collapse that afflicts the standard heuristic (e.g., recovering covariance traces close to their theoretical values where the median heuristic captures at most about half and often far less).

2 Mathematical background

We begin by considering a symmetric positive definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ and its associated RKHS $\mathcal{H}_0$. We define $\mathcal{H}$ as the d -fold Cartesian product of $\mathcal{H}_0$ equipped with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$ and the canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. Moreover, we denote by $x \cdot y$ the Euclidean inner product and $\nabla \cdot$ the divergence operator. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is denoted by $\mathcal{P}(\mathbb{R}^d)$ and $\mathcal{P}_p(\mathbb{R}^d)$ denotes the subspace of measures with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, we define the Wasserstein p -distance

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of couplings of μ and ν , i.e. the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [26] quantify the distance between two measures as the maximum deviation of integrals taken over a class of test functions—a worst-case comparison that becomes a concrete, computable quantity once the function class is fixed. To make this comparison feasible for measuring the distance of a sample to an intractable target distribution, Stein’s method [35] can be used to construct test functions which have zero mean w.r.t. the target. Indeed, for the operator $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable choice of kernel, we have $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [6, 20]. They defined the *kernelized Stein discrepancy* (KSD) as

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. This optimization problem is solved by $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_{\pi}^k d\mu$, where $\mathcal{A}_{\pi}^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$, and Φ_k is the feature map associated with the kernel k . As a result, the supremum evaluates to the RKHS norm of ψ , giving the equivalent characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and reference distribution μ_0 , SVGD transforms μ_0 into an approximation of π by choosing $\mu_{n+1} := T_{\sharp} \mu_n$, where $T_{\sharp} \cdot$ is the push-forward operator for the map $T(x) = x + \gamma \psi^{\mu_n}(x)$, with the vector field $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_{\pi}^k d\mu$ for $\mu \in \mathcal{P}(\mathbb{R}^d)$ being the direction of steepest descent. This is motivated by the fact that the solution of Equation (1) implies that $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ is the minimizer of $\frac{d}{d\gamma} \text{KL}(T_{\sharp} \mu_n \parallel \pi) \Big|_{\gamma=0}$ in the unit ball of $\mathcal{H}$ [cf. 18, Theorem 3.1], where $\text{KL}(\cdot \parallel \cdot)$ denotes the KL-divergence. In particular, for this choice we have

$$\frac{d}{d\gamma} \text{KL} \left((\text{Id} + \gamma \psi^{\mu_n})_{\sharp} \mu_n \parallel \pi \right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Iteratively applying the maps T generated in this way to a particle set $\{X_0^i\}_{i=1}^M$ sampled from μ_0 leads to the following particle updates:

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Since proving convergence of SVGD with respect to the KL divergence is challenging and requires restrictive assumptions, recent work has shifted attention to analyzing convergence in terms of the kernelized Stein discrepancy (KSD) [16, 33, 32]. However, minimizing the KSD alone does not guarantee weak convergence: the sequence of mean-field measures may fail to be tight, which is a necessary condition for weak convergence of probability measures in Polish spaces. Moreover, as seen from the optimization problem in Equation (1), the RKHS structure (and therefore the choice of kernel) directly determines how well convergence in KSD translates into weak convergence [11].

To address this issue and strengthen convergence guarantees, we formulate our adaptive variant of SVGD using a parameterized family of kernels $\{k_{\theta} \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, we denote by KSD_{θ} the corresponding *kernelized Stein discrepancy*, by ψ_{θ}^{μ} the optimal update direction, and by Φ_{θ} the associated feature map. This explicit parameterization allows us to adaptively select kernels during optimization. In doing so, we account for the kernel’s influence on convergence properties and mitigate the limitations of relying on a fixed kernel choice.

Our approach builds on the idea that it is advantageous to select a kernel that maximizes the KSD between the empirical particle distribution and the target distribution. The rationale is quantitative and direct: the instantaneous decrease in KL divergence under SVGD equals the squared KSD at the current particle measure (see Equations (2) and (7)). Choosing, at each iteration, the kernel that yields the largest KSD therefore maximizes the achievable rate of KL decrease among all kernels in the class—a greedy step-wise optimum rather than a heuristic. This perspective is reinforced by geometric analyses of SVGD as a gradient flow, which show that kernels

inducing larger KSD values yield strictly more favorable convergence properties when the associated RKHSs are compared [28, 8].

While vanilla SVGD usually aims to guarantee convergence of KSD for a fixed kernel, our proposed algorithm targets the worst-case KSD within the kernel class. As opposed to the median heuristic used by Liu and Wang [18], the approach is applicable to any class of parameterized kernels: At each step of the algorithm, we try to find an optimal parameter

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_{\theta}(\mu_n | \pi).$$

We implement this idea by adjusting the kernel parameter using one or possibly more gradient ascent steps with a step size $s > 0$ for KSD before executing the particle update and also allow the possibility of not updating the parameter at every step of SVGD to decrease runtime (see Algorithm 1). To enable this option, we introduce a user-specified decision variable `paramupdate`. It is worth noting that the base SVGD step for transporting the particles can be replaced by more advanced variants (e.g., adaptive step-size schedules, line searches, momentum, or second-order/preconditioned updates) without modifying our kernel-selection mechanism. Similarly, we may replace the gradient ascent on the kernel parameter with alternative optimization schemes. Our implementation of the kernel update is based on the formula [6, 20]

$$\operatorname{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_{\pi}^k d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_{\pi}^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \operatorname{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

from which the necessary gradient can be computed directly. Two points show that these gains come at essentially no computational premium over vanilla SVGD:

- The gradient ascent steps for the kernel parameter reuse exactly the same gradients $\nabla \log \pi(X_n^i)$ already required by the SVGD step, so Ad-SVGd incurs *no* additional gradient evaluations of $\log \pi$ —the dominant per-iteration cost in most applications.
- Any residual overhead can be driven down further by using a small number of kernel-update steps `nstepstheta` or by updating the kernel parameter only once every several SVGD steps. In our ODE experiment, for instance, updating the bandwidth once per 100 particle updates leaves no measurable runtime difference between Med-SVGd and Ad-SVGd. When the number of particles M is large, subsampled particles suffice to approximate the KSD empirically.

Algorithm 1: Ad-SVGD

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps $nsteps, nstepstheta \in \mathbb{N}$

Output: Final particle set $\{X_{nsteps}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```
for  $n = 0$  to  $nsteps - 1$  do
    Decide  $paramupdate \in \{\text{True}, \text{False}\}$ ;
    if  $paramupdate$  then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to  $nstepstheta - 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{nstepstheta}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end
```

4 Convergence analysis

We consider a target measure with Lebesgue density of the form $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. To place Ad-SVGD on firm theoretical footing, we extend recent KSD-based convergence results for SVGD to the adaptive setting. Concretely, we build on the analysis of [32], which operates under Talagrand’s inequality—a condition that holds under mild assumptions on the target and is strictly weaker than the commonly employed logarithmic Sobolev inequality—and we carry over exactly the same assumptions as [32], now imposed uniformly over all kernel parameters. This lets us state guarantees for the *worst-case* KSD over the whole kernel class rather than for a single fixed kernel.

Assumption 1. We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.

Assumption 2. We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand’s inequality $T1$, which means that there exists $\lambda > 0$ such that

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \parallel \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.

4.1 Convergence analysis of SVGD under Talagrand's inequality

In the following, we denote

$$\mathcal{F}(\mu) := \text{KL}(\mu \parallel \pi).$$

and make use of the following fundamental inequality. Given a fixed kernel parameter $\theta \in \Theta$ such that Assumption 3 is satisfied, define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, let $\gamma, B > 0$, $\alpha > 1$ and $g \in \mathcal{H}_\theta$ such that $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$. Then, according to Proposition 3.1 in [32], it holds

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta} \quad (6)$$

with $K = (\alpha^2 + L)B$. For the iterates (3) with a fixed kernel parameter $\theta \in \Theta$ one can then derive the descent condition [32, Theorem 3.2],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1}. \quad (8)$$

The key aspect to verify (7) is the verification of $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$ using Talagrand's inequality, Assumption 2, which allows to apply (6) for $g = \psi_\theta^{\mu_n}$. Condition (7) can then be used to argue that

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

since $\sum_{n=0}^{\infty} \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) > 0$.

When introducing an adaptive kernel parameter choice (θ_n) , the inequality (7) changes to

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

Suppose the adaptive SVGD iteration can be written in the following simplified form

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n}) \# \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi), \end{aligned} \quad (9)$$

meaning that

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

We emphasize that this formulation is possible when the maximization of the KSD with respect to the kernel parameter has a (unique) closed-form solution. This is actually the case in the

multiple-kernel SVGD framework [1], which parameterizes the kernel as a convex combination of base kernels

$$k_\theta(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

For the general setting, the kernel parameter update θ_n in (9) needs to be approximated. Note that in the general setup we assume that $\operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The following descent condition is a direct consequence of Theorem 3.2 in [32]:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

there exists $c_\gamma > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) = 0$.*

The situation changes when we assume that we can only approximately solve the maximization task of the KSD with respect to the kernel parameter. Suppose that $\max_{\theta \in \Theta} \operatorname{KSD}_\theta(\mu | \pi) < \infty$ for any $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider the alternating scheme

$$\begin{aligned} \theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\operatorname{Id} + \gamma \psi_{\theta_n}^{\mu_n})_\# \mu_n \end{aligned} \tag{10}$$

for some sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ with the goal to maximize $\operatorname{KSD}_\theta(\mu_n | \pi)$. Specifically, in Algorithm 1 Ψ_n corresponds to the gradient ascent update for the KSD. However, as mentioned above, this update could be replaced by other suitable iterative optimization schemes. Our required assumption on the update rule is the following convergence behavior.

Assumption 4. *We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and*

$$\max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) - \operatorname{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

Using this assumption, we can make the following convergence guarantee.

Theorem 3. *Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_\theta(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [32] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \text{KSD}_{\theta_n}^2(\mu_n|\pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and, in particular, it is a Cauchy sequence. Hence, we have

$$c_\gamma \text{KSD}_{\theta_n}^2(\mu_n|\pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi) \leq \text{KSD}_{\theta_n}^2(\mu_n|\pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under the so-called Stein logarithmic Sobolev inequality, we can strengthen these guarantees from convergence of the KSD to convergence in the KL divergence itself, and, under specific decay scenarios for ε_n , obtain explicit iteration-complexity rates for the mean-field limiting system (10). This same setting quantifies exactly how much bias remains when the KSD maximization is solved only up to a fixed $\bar{\varepsilon}$ -accuracy per iteration, so the price of approximate kernel optimization is bounded in closed form; the details are given in Section B. Crucially, this condition need hold for only a *single* kernel in our class $\{k_\theta\}$ —a substantially weaker and more easily verifiable requirement than asking it of a fixed kernel, and one that the adaptive maximization is designed to exploit.

5 Numerical Experiments

We now evaluate Ad-SVGD empirically, in every case comparing it against SVGD with the median heuristic (Med-SVGD) under identical dynamics, step sizes, and particle budgets so that kernel selection is the only variable. Two experiments are presented in the main text: a one-dimensional Gaussian mixture model, which isolates bandwidth sensitivity, and a linear inverse problem governed by an ordinary differential equation, which exposes variance collapse. Controlled scaling studies that trace how the gap between the two methods widens with dimension are provided in Section A, and Section 5.4 reports a Bayesian logistic regression benchmark used in prior SVGD literature on which Ad-SVGD attains both higher prediction accuracy and a smaller discrepancy to an MCMC reference.

5.1 Kernel parameterization

SVGD is most commonly used with kernels of the form

$$k_h(x, y) = \exp\left(-\frac{\|x - y\|_p^p}{h}\right),$$

where $\|\cdot\|_p$ denotes the p -norm on $\mathbb{R}^d$ [e.g. 18, 2, 8]. We will focus on selection strategies for the parameter h , which is known as the kernel *bandwidth*. The commonly used heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, where med denotes the current median distance between the particles. This choice is motivated by the goal of balancing the two terms in the SVGD update (3) [18].

To exploit the flexibility of our adaptive method, we use product kernels of the form

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with parameter $h = (h_1, \dots, h_d)$, i.e. a separate bandwidth per dimension—exactly the degrees of freedom the scalar median heuristic lacks. The derivatives needed to apply Algorithm 1 with these kernels (see Equations (4) and (5)) are available in closed form. For a fair comparison we also tested a per-dimension median heuristic, taking a naive median in each coordinate; this variant did not close the gap and still suffered from variance collapse, indicating that the advantage of Ad-SVGD comes from actively maximizing the KSD rather than merely from having more bandwidth parameters.

Beyond the kernels used here, the same adaptive mechanism extends to more scalable kernel families in higher-dimensional settings. In particular, spectral or low-rank parametrizations can adapt the dominant directions of variability without the exponential parameter growth of full product kernels, while Matérn or mixed-product constructions add explicit control over smoothness and anisotropy—avenues that the consistent margins reported below make worth pursuing.

5.2 Toy example

We first consider the one-dimensional example from Liu and Wang [18], a two-component Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. Starting from an initial set of $M \in \{50, 200, 500\}$ particles sampled from $\mathcal{N}(0, 1)$, we run SVGD for 10^4 steps with step size 1, using the kernels described above with $p = 1$ and a range of fixed bandwidths h . Sample quality is measured by the Wasserstein-1 distance $\mathcal{W}_1$, computed from the explicit formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see Panaretos and Zemel [29]) against an exact sample of size 10^5 ; this gives a directly comparable error for every bandwidth. The outcome, in Figure 1, is stark: low error is confined to a narrow window of h , and the final $\mathcal{W}_1$ distance swings by more than an order of magnitude as h moves outside its band. This sharp, quantitative sensitivity is exactly the failure mode that a principled selection strategy must remove, and it motivates the adaptive scheme we evaluate next.

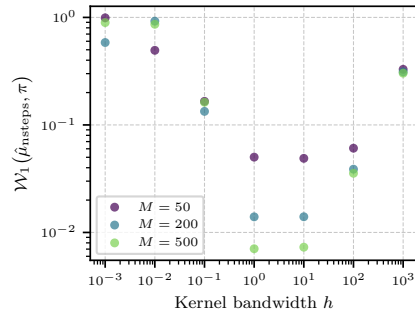


Figure 1: Bandwidth sensitivity on the one-dimensional Gaussian mixture: final Wasserstein-1 distance to the target for SVGD run with a range of fixed bandwidths h . Only a narrow band of h yields low error; outside it, the distance grows by an order of magnitude as h moves outside its band. This sharp, quantitative sensitivity is exactly the failure mode that a principled selection strategy must remove, and it motivates the adaptive scheme we evaluate next.

5.3 Linear inverse problem based on ODE

The following example is adapted from [38, Example 2.1]. We consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

and the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete noisy observation points of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. These observations are described by

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ is observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is defined by $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ being the observation operator at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation of the inverse problem, we consider a Gaussian process (GP) prior for u given by the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and $x_k \sim \mathcal{N}(0, \lambda_k)$ independently with $\lambda_k = 50k^{-2}$. The resulting problem is to estimate the KL coefficients $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ with prior given by $\mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. Assuming $\varepsilon \sim \mathcal{N}(0, \Gamma)$, this leads to the posterior mean, but only Ad-SVGD reproduces the width of the true posterior band; Med-SVGD visibly collapses the uncertainty.

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|_2^2 - \frac{1}{2} \|x\|_{\Gamma_0}^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

For the implementation, we replace H by a numerical discretization operator for Equation (11) using a grid with mesh size 2^{-8} and consider the fully observed system (i.e. $N_{\text{obs}} = 2^8$). We use $N_x = 16$ terms for the KL expansion of u and assume noise covariance $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. We construct reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We compare SVGD with the median heuristic (Med-SVGD) against Ad-SVGD across particle ensemble sizes $M \in \{50, 100, 200\}$, using the kernels of Section 5.1 with $p = 1$. Both algorithms are run under identical conditions— $4 \cdot 10^5$ iterations, particle step size 10^{-3} , and, following Liu and Wang [18], a variant of AdaGrad for adaptive step-size control—so that only the kernel-selection strategy differs. Ad-SVGD updates the bandwidth with step size 10^{-5} , and only once per 100 particle updates; under this schedule the two methods show no significant runtime difference, so any accuracy gap is obtained essentially for free.

For a single representative seed, Figure 2 shows that both methods recover the posterior mean, but only Ad-SVGD reproduces the posterior uncertainty—Med-SVGD collapses the confidence band. To turn this into a quantitative comparison, we use the Wasserstein-2 distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the Gaussian posterior π and the fitted normal $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, with $\hat{\mu}$ the sample mean and $\hat{\Sigma}$ the sample covariance of the particle set [29, Section 3.2]. Because π is multivariate normal, this admits the explicit form

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π are the posterior mean and covariance, and we additionally compare the per-component marginal variances against the posterior. Aggregated over 56 random seeds (the posterior covariance is independent of $\bar{y}$), Figure 3 shows a consistent ordering: Ad-SVGD attains a

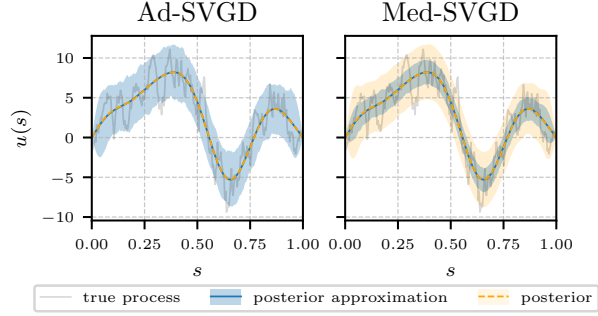
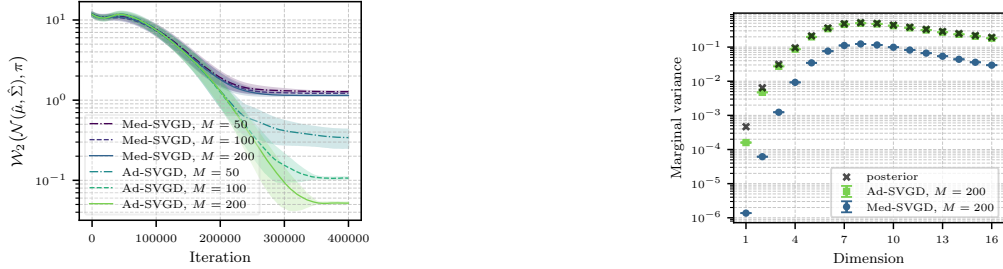


Figure 2: GP reconstruction for the ODE-based inverse problem (mean and 90% confidence interval). Both methods recover the posterior mean, but only Ad-SVGD reproduces the width of the true posterior band; Med-SVGD visibly collapses the uncertainty.

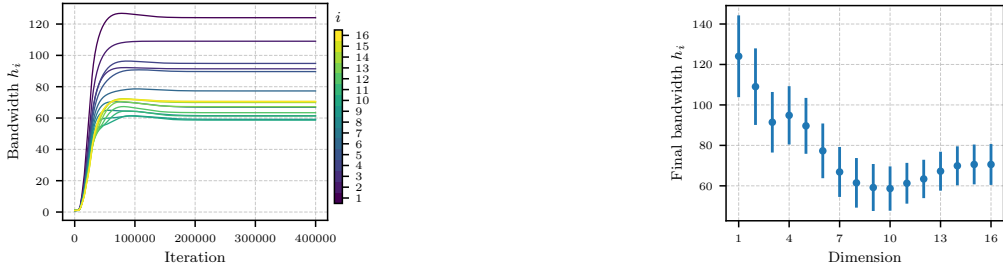
lower $\mathcal{W}_2$ error than Med-SVGD, whose collapse of the marginal variances is the dominant source of its error. The two methods also respond differently to more particles—Ad-SVGD’s approximation improves as M grows beyond 50, whereas Med-SVGD stagnates, so the gap widens rather than closes with additional compute. Figure 4 explains the mechanism: the component-wise bandwidths learned by Ad-SVGD stabilize well before the approximation error does, settle at clearly distinct values across components, and are negatively correlated with the corresponding marginal variances—i.e. the method automatically assigns smaller bandwidths to higher-variance directions.



(a) Convergence of the $\mathcal{W}_2$ approximation error: Ad-SVGD converges to a lower error, and improves with more particles, whereas Med-SVGD plateaus.

(b) Marginal variances: Ad-SVGD tracks the posterior variances closely, while Med-SVGD underestimates them.

Figure 3: ODE-based inverse problem, Med-SVGD vs. Ad-SVGD (mean and 95% confidence interval over 56 random seeds). Across both metrics Ad-SVGD is consistently closer to the true posterior.



(a) Per-dimension bandwidth evolution (mean): bandwidths stabilize well before the approximation error does.

(b) Final bandwidths (mean and 95% conf. interval): clearly distinct across components and negatively correlated with the marginal variances.

Figure 4: Bandwidths learned by Ad-SVGD on the ODE-based inverse problem, aggregated over 56 random seeds. The learned, component-specific values are what allow Ad-SVGD to avoid the variance collapse seen in Figure 3.

5.4 Bayesian logistic regression

As in Liu and Wang [18] and Liu et al. [21], we consider a Bayesian logistic regression (BLR) model applied to the Covtype data set [4]. For the regression weights w , we assign a Gaussian prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$ and infer the posterior of $x = [w, \alpha]$.

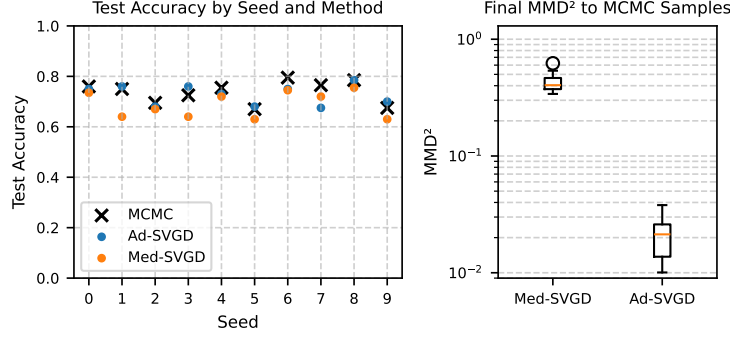


Figure 5: Bayesian logistic regression, Med-SVGD vs. Ad-SVGD over 10 random data draws: prediction accuracy (left, higher is better) and MMD^2 to the NUTS reference samples (right, lower is better). Ad-SVGD is simultaneously more accurate and closer to the reference posterior on almost every draw.

This benchmark lets us compare the two methods on both a downstream task metric (prediction accuracy) and a distributional metric (discrepancy to a reference posterior). Following Liu et al. [21], we generate that reference with the No-U-Turn Sampler (NUTS) [13], giving a common yardstick against which both Med-SVGD and Ad-SVGD are scored. Again following Liu et al. [21], we test the methods on size-1000 subsets of the data set, aggregating over 10 random draws. Classification performance is measured by the test-set prediction accuracy of the particle mean. The comparison in Figure 5 is one-sided: Ad-SVGD attains the higher test accuracy on nearly every one of the 10 draws, and does so while simultaneously sitting closer to the MCMC reference—so the accuracy gain does not come at the expense of posterior fidelity.

The same figure quantifies posterior fidelity via the squared maximum mean discrepancy (MMD^2) [25, Section 3.5] to the MCMC samples, and here the gap between the two methods is pronounced. Figure 7 makes its origin explicit: comparing the final particle covariance matrices to the MCMC reference (averaged over the 10 seeds, following the visualization of Liu et al. [21]), Med-SVGD severely underestimates the posterior uncertainty, whereas Ad-SVGD reproduces the reference covariance structure far more faithfully. The learned bandwidths in Figure 6 account for this: starting from an uninformed initialization (1 in every component), Ad-SVGD recovers component-specific bandwidths spanning several orders of magnitude, which is precisely the adaptivity the median heuristic cannot supply.

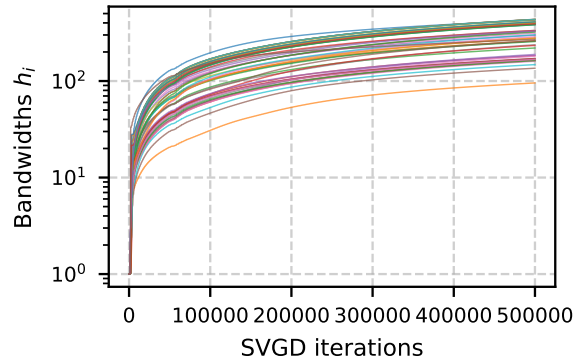


Figure 6: Bandwidth evolution for the BLR example: from an uninformed initialization of 1 in every component, Ad-SVGD drives the per-component bandwidths to values differing by several orders of magnitude.

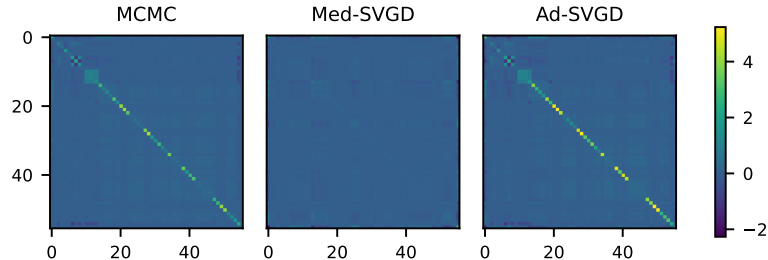


Figure 7: Final particle covariance matrices for Med-SVGD and Ad-SVGD compared to the MCMC reference (mean over 10 seeds). Ad-SVGD closely matches the reference covariance, whereas Med-SVGD collapses toward near-zero off-diagonal and diagonal magnitudes, visibly underestimating the posterior uncertainty.

6 Limitations and outlook

The main limitation of our analysis is its reliance on Assumption 4, which we do not prove holds for the alternating gradient-ascent scheme of Algorithm 1; establishing when it does is the subject of ongoing work. We stress, however, that this is a gap in the theory, not in the observed behavior: the very implementation whose convergence we cannot yet certify is the one that delivers the consistent, dimension-robust accuracy gains reported throughout Section A and Section 5.4.

Because our analysis and experiments deliberately keep the original SVGD dynamics fixed—so that the reported margins can be attributed to kernel selection alone—an immediate next step is to combine adaptive kernel selection with recent SVGD variants such as sliced SVGD [10], Grassmann SVGD [21], or Stein transport [27], where the two sources of improvement may compound.

References

- [1] Qingzhong Ai et al. “Stein Variational Gradient Descent with Multiple Kernels”. In: *Cognitive Computation* 15.2 (2023), pp. 672–682. DOI: [10.1007/s12559-022-10069-5](https://doi.org/10.1007/s12559-022-10069-5).
- [2] Jimmy Ba et al. “Understanding the Variance Collapse of SVGD in High Dimensions”. In: *International Conference on Learning Representations*. 2022.
- [3] Krishna Balasubramanian, Sayan Banerjee, and Promit Ghosal. “Improved Finite-Particle Convergence Rates for Stein Variational Gradient Descent”. In: *The Thirteenth International Conference on Learning Representations*. 2025.
- [4] Jock Blackard. *Coverttype*. UCI Machine Learning Repository. 1998. DOI: [10.24432/C50K5N](https://doi.org/10.24432/C50K5N).
- [5] Sinho Chewi et al. “SVGD as a kernelized Wasserstein gradient flow of the chi-squared divergence”. In: *Advances in Neural Information Processing Systems* 33. 2020.
- [6] Kacper Chwialkowski, Heiko Strathmann, and Arthur Gretton. “A Kernel Test of Goodness of Fit”. In: *Proceedings of The 33rd International Conference on Machine Learning*. 2016.
- [7] Gianluca Detommaso et al. “A Stein variational Newton method”. In: *Advances in Neural Information Processing Systems* 31. 2018.
- [8] Andrew Duncan, Nikolas Nüsken, and Lukasz Szpruch. “On the geometry of Stein variational gradient descent”. In: *Journal of Machine Learning Research* 24 (2023).
- [9] Rémi Flamary et al. “POT: Python Optimal Transport”. In: *Journal of Machine Learning Research* 22 (2021).
- [10] Wenbo Gong, Yingzhen Li, and José Miguel Hernández-Lobato. “Sliced Kernelized Stein Discrepancy”. In: *International Conference on Learning Representations*. 2021.
- [11] Jackson Gorham and Lester Mackey. “Measuring Sample Quality with Kernels”. In: *Proceedings of the 34th International Conference on Machine Learning*. 2017.
- [12] Arthur Gretton et al. “A Kernel Two-Sample Test”. In: *Journal of Machine Learning Research* 13 (2012).
- [13] Matthew D. Hoffman and Andrew Gelman. “The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo”. In: *Journal of Machine Learning Research* 15 (2014).
- [14] Richard Jordan, David Kinderlehrer, and Felix Otto. “The Variational Formulation of the Fokker–Planck Equation”. In: *SIAM Journal on Mathematical Analysis* 29.1 (1998), pp. 1–17. DOI: [10.1137/S0036141096303359](https://doi.org/10.1137/S0036141096303359).
- [15] Rahif Kassab and Osvaldo Simeone. “Federated Generalized Bayesian Learning via Distributed Stein Variational Gradient Descent”. In: *IEEE Transactions on Signal Processing* 70 (2022), pp. 2180–2192. DOI: [10.1109/TSP.2022.3168490](https://doi.org/10.1109/TSP.2022.3168490).
- [16] Anna Korba et al. “A Non-Asymptotic Analysis for Stein Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems* 33. Vol. 33. 2020.
- [17] Lauro Langosco di Langosco, Vincent Fortuin, and Heiko Strathmann. “Neural Variational Gradient Descent”. In: *Fourth Symposium on Advances in Approximate Bayesian Inference*. 2022.

- [18] Q. Liu and D. Wang. “Stein Variational Gradient Descent: A General Purpose Bayesian Inference Algorithm”. In: *Advances in Neural Information Processing Systems 29*. 2016.
- [19] Qiang Liu. “Stein Variational Gradient Descent as Gradient Flow”. In: *Advances in Neural Information Processing Systems 30*. 2017.
- [20] Qiang Liu, Jason Lee, and Michael Jordan. “A Kernelized Stein Discrepancy for Goodness-of-fit Tests”. In: *Proceedings of The 33rd International Conference on Machine Learning*. 2016.
- [21] Xing Liu et al. “Grassmann Stein Variational Gradient Descent”. In: *Proceedings of The 25th International Conference on Artificial Intelligence and Statistics*. 2022.
- [22] Yang Liu et al. “Stein Variational Policy Gradient”. In: *Proceedings of the Thirty-Third Conference on Uncertainty in Artificial Intelligence*. 2017.
- [23] Jianfeng Lu, Yulong Lu, and James Nolen. “Scaling Limit of the Stein Variational Gradient Descent: The Mean Field Regime”. In: *SIAM Journal on Mathematical Analysis* 51.2 (2019), pp. 648–671. DOI: [10.1137/18M1187611](https://doi.org/10.1137/18M1187611).
- [24] Safa Messaoud et al. “S2AC: Energy-Based Reinforcement Learning with Stein Soft Actor Critic”. In: *The Twelfth International Conference on Learning Representations*. 2024.
- [25] Krikamol Muandet et al. “Kernel Mean Embedding of Distributions: A Review and Beyond”. In: *Foundations and Trends in Machine Learning* 10.1-2 (2017), pp. 1–141. DOI: [10.1561/22000000060](https://doi.org/10.1561/22000000060).
- [26] Alfred Müller. “Integral Probability Metrics and Their Generating Classes of Functions”. In: *Advances in Applied Probability* 29.2 (1997), pp. 429–443. DOI: [10.2307/1428011](https://doi.org/10.2307/1428011).
- [27] Nikolas Nüsken. *Stein transport for Bayesian inference*. 2024. arXiv: [2409.01464](https://arxiv.org/abs/2409.01464).
- [28] Nikolas Nüsken and D. R. Michiel Renger. “Stein variational gradient descent: Many-particle and long-time asymptotics”. In: *Foundations of Data Science* 5.3 (2023), pp. 286–320. DOI: [10.3934/fods.2022023](https://doi.org/10.3934/fods.2022023).
- [29] Victor M. Panaretos and Yoav Zemel. “Statistical Aspects of Wasserstein Distances”. In: *Annual Review of Statistics and Its Application* 6 (2019), pp. 405–431. DOI: [10.1146/annurev-statistics-030718-104938](https://doi.org/10.1146/annurev-statistics-030718-104938).
- [30] Yuchen Pu et al. “VAE Learning via Stein Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems 30*. 2017.
- [31] Sebastian Reich and Simon Weissmann. “Fokker–Planck Particle Systems for Bayesian Inference: Computational Approaches”. In: *SIAM/ASA Journal on Uncertainty Quantification* 9.2 (2021), pp. 446–482. DOI: [10.1137/19M1303162](https://doi.org/10.1137/19M1303162).
- [32] Adil Salim, Lukang Sun, and Peter Richtarik. “A Convergence Theory for SVGD in the Population Limit under Talagrand’s Inequality T1”. In: *Proceedings of the 39th International Conference on Machine Learning*. 2022.
- [33] Jiaxin Shi and Lester Mackey. “A Finite-Particle Convergence Rate for Stein Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems 36*. 2023.
- [34] Björn Sprungk, Simon Weissmann, and Jakob Zech. “Metropolis-adjusted interacting particle sampling”. In: *Statistics and Computing* 35.64 (2025). DOI: [10.1007/s11222-025-10595-w](https://doi.org/10.1007/s11222-025-10595-w).

- [35] Charles Stein. “A bound for the error in the normal approximation to the distribution of a sum of dependent random variables”. In: *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, Volume 2: Probability Theory*. 1972, pp. 583–603.
- [36] Dilin Wang, Zhe Zeng, and Qiang Liu. “Stein Variational Message Passing for Continuous Graphical Models”. In: *Proceedings of the 35th International Conference on Machine Learning*. 2018.
- [37] Dilin Wang et al. “Stein Variational Gradient Descent With Matrix-Valued Kernels”. In: *Advances in Neural Information Processing Systems 32*. 2019.
- [38] Simon Weissmann, Ashia Wilson, and Jakob Zech. “Multilevel Optimization for Inverse Problems”. In: *Proceedings of Thirty Fifth Conference on Learning Theory*. 2022.
- [39] Qian Zhao et al. “Stein variational gradient descent with learned direction”. In: *Information Sciences* 637 (2023). DOI: [10.1016/j.ins.2023.118975](https://doi.org/10.1016/j.ins.2023.118975).
- [40] Jingwei Zhuo et al. “Message Passing Stein Variational Gradient Descent”. In: *Proceedings of the 35th International Conference on Machine Learning*. 2018.

A More numerical experiments

A.1 Gaussian mixture models

We revisit the example from Section 5.2. For the same setup, Figure 9 contrasts histograms of the final particle distribution with the target density for three fixed bandwidths $h \in \{0.001, 1, 1000\}$, illustrating concretely how far off a poorly chosen fixed bandwidth can be. We then compare Med-SVGD and Ad-SVGD directly, examining the final particle distributions after 10^4 iterations with step size 1 across a range of particle counts.

Figure 8 reports the Wasserstein-1 distance to the target for $M = 10, 20, 50, 100, 200, 500$. As expected, essentially the same error, reaching below 0.01 at $M = 500$ throughout, though reaching a dimension in this benign one-dimensional regime the median heuristic is already near-optimal, so Ad-SVGD matches rather than beats it—the setting where we should expect no advantage, and observe none.

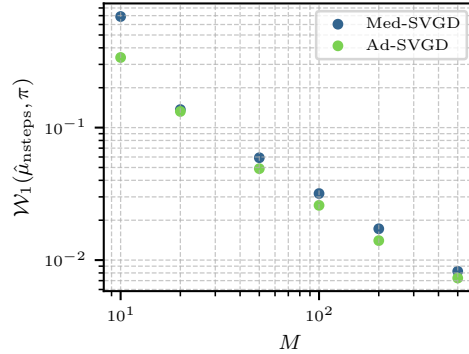


Figure 8: Approximation error (Wasserstein-1) vs. particle count M for the 1D Gaussian mixture. Both methods im-

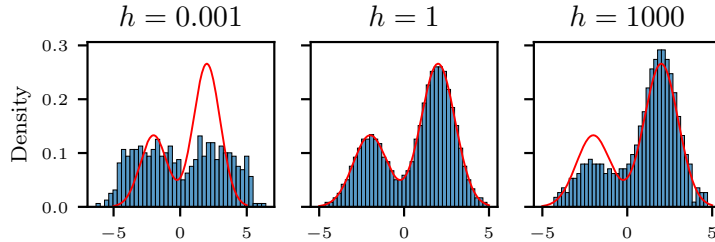


Figure 9: Final particle histograms for fixed bandwidths $h = 0.001, 1, 1000$ (target density π in red). The intermediate bandwidth $h = 1$ matches π well, while the extreme choices $h = 0.001$ and $h = 1000$ produce visibly wrong distributions—a direct picture of the bandwidth sensitivity quantified in Figure 1.

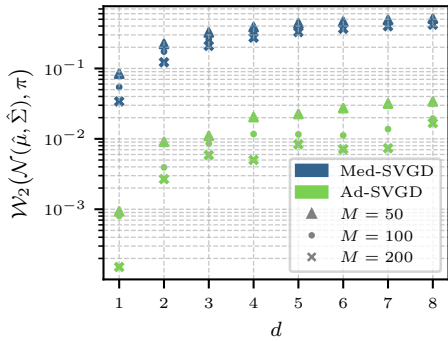
In summary, in one dimension both methods approximate the target well and are statistically indistinguishable in accuracy: adaptive bandwidth selection performs on par with the median heuristic without any advantage here. This is the expected baseline outcome, and it sets up the contrast with the higher-dimensional experiments below, where the same adaptive mechanism begins to separate decisively from the heuristic.

A.2 Scaling dimension: multivariate normal distribution

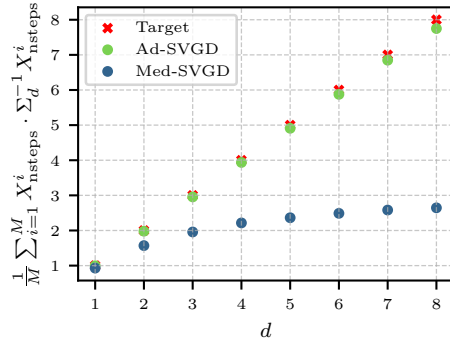
Moving into higher dimensions, we now consider the Gaussian targets $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$ —a controlled sweep in which the only thing that changes is

the dimension. The initial particles are drawn from $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$ and both algorithms run for 10^4 iterations with step size 0.1 (reduced, with a correspondingly increased iteration count, only where numerical stability required).

We track how approximation quality degrades with dimension for each method. Since the Wasserstein-2 distance between two Gaussians is available in closed form (see [29]), we score sample quality by the Wasserstein-2 distance between π_d and the fitted Gaussian $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, with $\hat{\mu}, \hat{\Sigma}$ the empirical mean and covariance of the particles [29, Section 3.2]; following the Python Optimal Transport library [9] used for the computation, we refer to this as the *Bures Wasserstein distance*. Figure 10a plots this measure for $M = 50, 100, 200$ as d increases. The separation is unambiguous: Ad-SVGD attains a lower Bures Wasserstein distance than Med-SVGD at *every* dimension $d \in \{2, \dots, 8\}$ and every particle count, and the margin grows with d .



(a) Bures Wasserstein error vs. dimension d for $M \in \{50, 100, 200\}$. Ad-SVGD is lower at every d , with the gap over Med-SVGD widening as d grows.



(b) Mean χ^2 statistic for $M = 200$ (true expectation = d , dashed). Ad-SVGD stays near d across all dimensions, while Med-SVGD falls progressively below it.

Figure 10: Multivariate normal targets of increasing dimension $d = 2, \dots, 8$: on both the Bures Wasserstein error and the χ^2 variance diagnostic, Ad-SVGD tracks the truth while Med-SVGD deteriorates as d grows.

The Bures Wasserstein distance is relatively insensitive to the small-variance trailing components of π_d ; to expose how badly the uncertainty in those components is captured, we turn to the statistic $X \cdot \Sigma_d^{-1} X$, which for $X \sim \mathcal{N}(0, \Sigma_d)$ is χ^2 -distributed with d degrees of freedom and hence has expected value exactly d (cf. [34, p. 27]). Comparing the particle mean of this statistic ($M = 200$) against the target value d in Figure 10b yields a clean separation: Med-SVGD’s statistic falls further below d with each increase in dimension—a direct, quantitative signature of variance collapse—whereas Ad-SVGD’s stays close to d throughout. The diagnostic thus pinpoints where Med-SVGD fails in higher dimensions and where Ad-SVGD does not.

Tables 1 and 2 make this component-by-component: for each d they list the achieved marginal variances alongside the target variances (top row), for Med-SVGD and Ad-SVGD respectively. The contrast is striking. Ad-SVGD matches every target variance to within a few percent even at $d = 8$ (e.g. 0.9691 vs. 1.0000 in the first component and 0.0150 vs. 0.0156 in the eighth), whereas Med-SVGD recovers less than half of the leading variance at $d = 8$ (0.4753 vs. 1.0000) and underestimates every component substantially. This mirrors the χ^2 diagnostic in Figure 10b exactly. Finally, we zoom in on the marginal distributions for $M = 200$ and $d = 8$. For visualization

Table 1: Marginal variances achieved by Med-SVGD ($M = 200$) for each dimension d , against the target marginal variances (top row). Every entry falls well below its target—e.g. the leading component drops from the target 1.0000 to 0.4753 at $d = 8$ —documenting a systematic underestimation of uncertainty that worsens with d . Compare with Table 2.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances achieved by Ad-SVGD ($M = 200$) for each dimension d , against the target marginal variances (top row). Values stay within a few percent of the target in every component and at every dimension—e.g. 0.9691 vs. 1.0000 (component 1) and 0.0150 vs. 0.0156 (component 8) at $d = 8$ —in sharp contrast to the collapse in Table 1.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

we rescale each component by the inverse target marginal standard deviation (multiplying the i -th component by i), which maps every marginal of π_d onto a standard normal. Figures 11 and 12 show the resulting normalized marginal histograms for Med-SVGD and Ad-SVGD (standard Gaussian overlaid for reference): Ad-SVGD reproduces all eight marginals, while Med-SVGD’s are visibly too narrow. The Q-Q plots in Figures 13 and 14, which compare the normalized marginal quantiles against those of a standard normal, tell the same story—Ad-SVGD’s points lie along the reference line, Med-SVGD’s bend away from it.

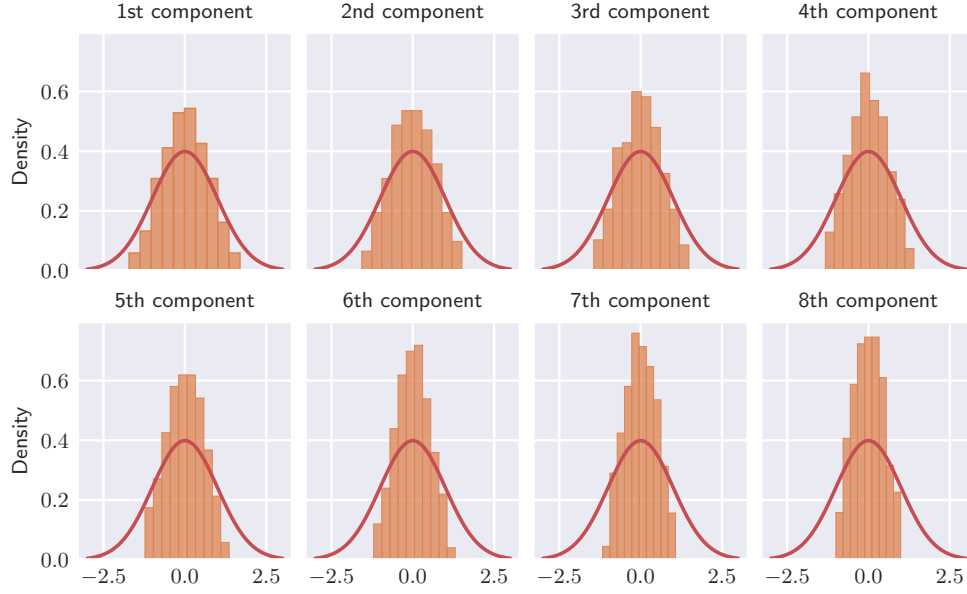


Figure 11: Normalized marginal histograms ($d = 8$, $M = 200$, Med-SVGd), rescaled so each target marginal is standard normal (overlaid). Every marginal is markedly too narrow, the visual counterpart of the collapsed variances in Table 1.

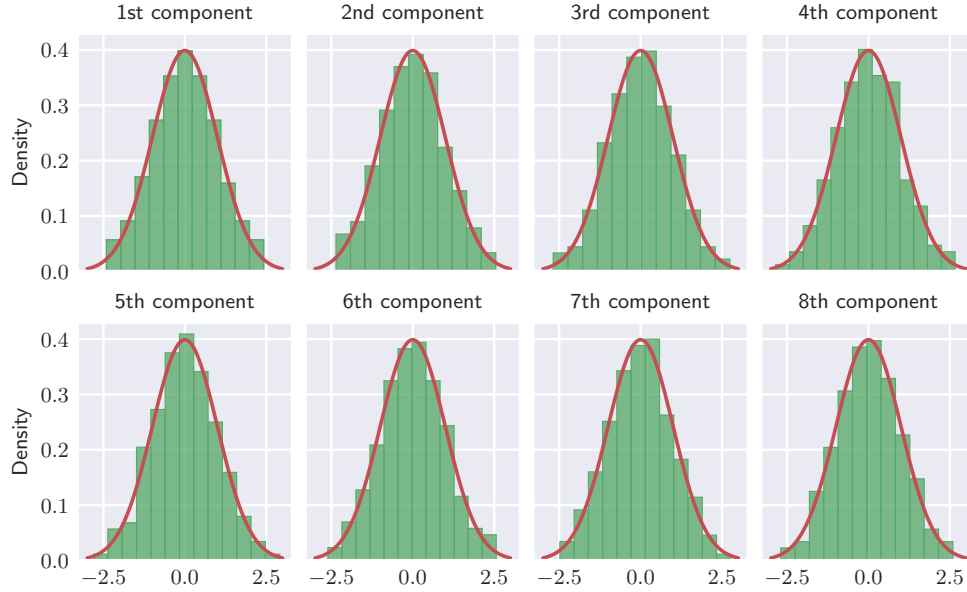


Figure 12: Normalized marginal histograms ($d = 8$, $M = 200$, Ad-SVGd), rescaled so each target marginal is standard normal (overlaid). All eight marginals closely match the reference density, consistent with the near-exact variances in Table 2.

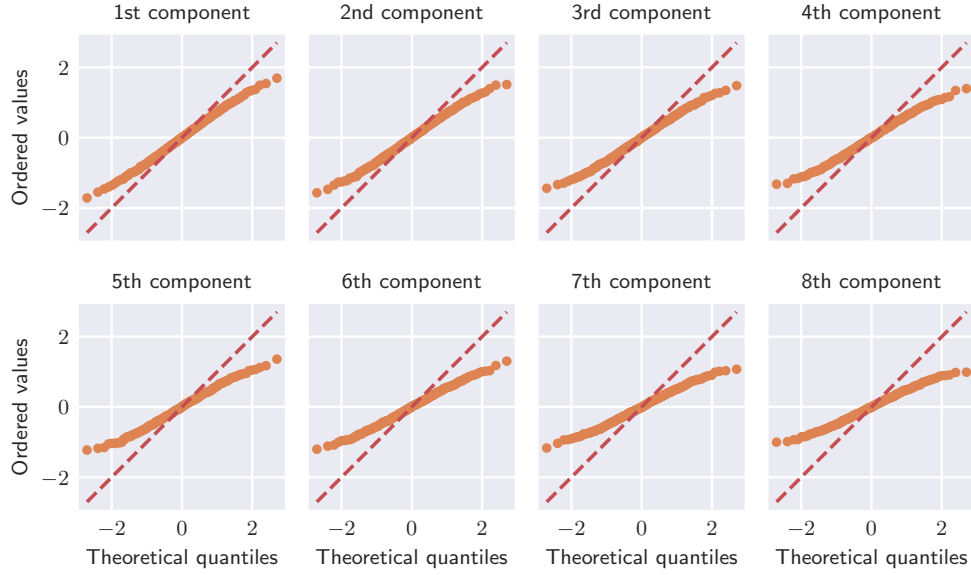


Figure 13: Q-Q plots of the normalized marginals ($d = 8$, $M = 200$, Med-SVGd) against a standard normal (reference line of slope 1 through the origin). The points bend systematically off the line, signaling under-dispersed marginals.

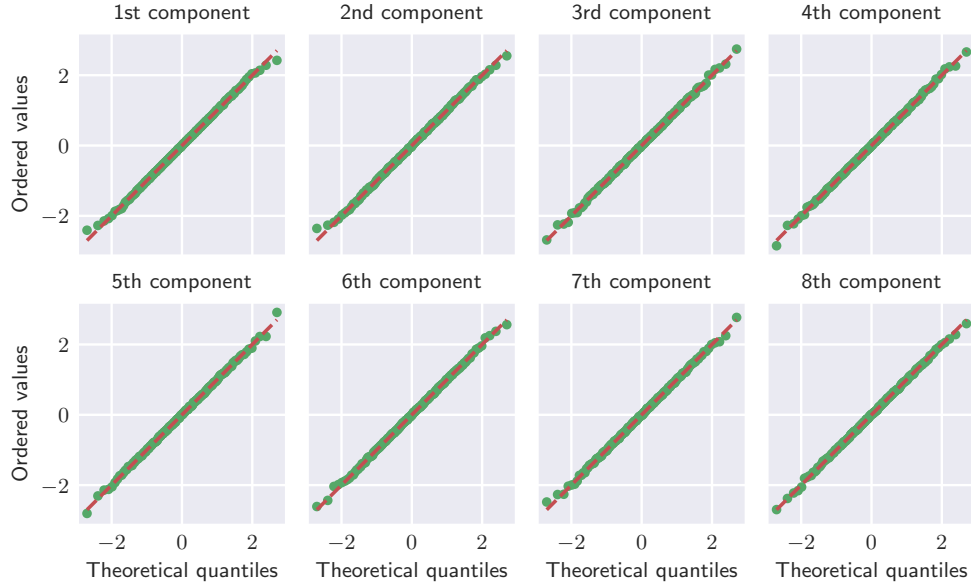


Figure 14: Q-Q plots of the normalized marginals ($d = 8$, $M = 200$, Ad-SVGd) against a standard normal (reference line of slope 1 through the origin). The points lie essentially on the reference line, confirming that all marginals are captured accurately.

A.3 Gaussian process inference

As a further scaling study, we evaluate Ad-SVGD on the Gaussian-process (GP) inference task proposed in [31], which lets us vary both the latent dimension N_x and the number of observations N_y and read off the effect on posterior-variance recovery. We consider a GP on $[0, 1]$ represented by a truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ is a vector of independent Gaussian random variables $x_k \sim \mathcal{N}(0, k^{-2})$. We observe the process at N_y equispaced points in $[0, 1]$ and infer the coefficients x_k . For fixed N_x and N_y , this corresponds to an inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$, $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal matrix Σ with entries k^{-2} , $k = 1, \dots, N_x$ and we assume independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. We construct reference observations $\bar{y} \in \mathbb{R}^{N_y}$ by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

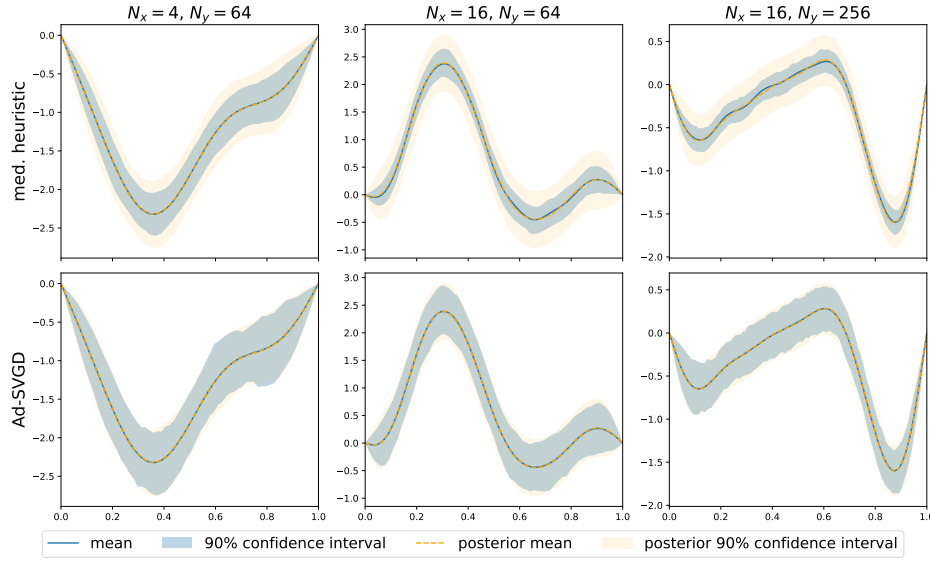


Figure 15: Estimated processes from the median heuristic and Ad-SVGD ($M = 100$) against the true posterior, for three configurations of (N_x, N_y) . As N_x grows the median heuristic’s uncertainty band shrinks below the posterior, while Ad-SVGD stays aligned with it across all observation counts.

We use SVGD to sample from the resulting posterior $X | Y = \bar{y}$ and compare the performance of the median heuristic with our adaptive approach. For SVGD with the median heuristic, we use kernels of the form $k(x, y) = \exp(-\|x - y\|_1/h)$; for Ad-SVGD, we use product kernels $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with parameter $h = (h_1, \dots, h_{N_x})$, i.e. we use a different bandwidth for each dimension. Following [18], we use a variant of Adagrad for step size control in the scenarios with $N_x = 16$.

Figure 15 overlays the estimated processes on the posterior for $(N_x, N_y) = (4, 64)$, $(16, 64)$, and $(16, 256)$. The pattern repeats: as N_x increases the median heuristic underestimates the posterior variance, whereas Ad-SVGD tracks it closely, and this holds consistently across the different observation counts. Table 3 makes the comparison numerical by reporting the trace of the particle covariance against the theoretical posterior trace (averaged over 25 runs). The gap is large and systematic: the median heuristic recovers roughly a quarter to a half of the true trace in every

configuration (e.g. 0.022 vs. a theoretical 0.086 at $N_x = 16, N_y = 64$), while Ad-SVGD lands close to the theoretical value in every column—within roughly 15% throughout and nearly exact for $N_x = 4$ (0.055 vs. 0.056)—confirming that the accurate variance recovery seen elsewhere persists under joint scaling of dimension and observations.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the final particle covariance ($M = 100$) versus the theoretical posterior trace for five (N_x, N_y) configurations, averaged over 25 runs. Ad-SVGD stays close to the theoretical trace in every column (within about 15%), while the median heuristic recovers only about a quarter to a half of it.

B Refined theoretical analysis

We can sharpen our convergence result—from decay of the KSD to an explicit, quantitative rate on the KL divergence—under a so-called Stein log-Sobolev inequality relating the two. Duncan, Nüsken, and Szpruch [8] introduced this inequality and discussed conditions under which it might hold. A notable advantage of the adaptive setting is that we may relax this requirement to just a *single* kernel in the entire class satisfying the log-Sobolev inequality, rather than a fixed one, which makes the condition easier to certify. We make this precise as follows.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \parallel \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

This property lets us quantify convergence directly in the KL divergence, with explicit rates. Notably, it yields a finite error bound on $\mathcal{F}(\mu_n)$ even under only an approximate condition $\varepsilon_n \leq \bar{\varepsilon}$, without requiring $\varepsilon_n \rightarrow 0$: the residual error is controlled by, and scales linearly with, the achievable KSD-maximization accuracy $\bar{\varepsilon}$.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n \mid \pi) - \text{KSD}_{\theta_n}^2(\mu_n \mid \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_{\gamma} \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_{\varepsilon} q^n$ for some $q \in (0, 1)$, $c_{\varepsilon} > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_{\varepsilon}}{(n+1)^p}$ for $c_{\varepsilon}, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_{\gamma} c_{\varepsilon}}{n^p} \frac{1}{1 - \rho} + c_{\gamma} c_{\varepsilon} \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. Similar to the proof of Theorem 3, under Assumption 5 and the conditions on α, γ , we have

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_{\gamma} \max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n \mid \pi) + c_{\gamma} \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_{\gamma} \varepsilon_n, \end{aligned}$$

where $c_{\gamma} = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$, we deduce the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_{\gamma} \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We consider the three different cases separately:

1. Under $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows by the computation of the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. Assuming that $\varepsilon_n \leq c_{\varepsilon} q^n$ for some $q \in (0, 1)$ and $c_{\varepsilon} > 0$ we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_{\gamma} c_{\varepsilon} \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

where we have used the formula for geometric series. Similarly, in the case of $q < \rho$, we can directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, in the case $q = \rho$, we simply bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and deduce the claim line by line as before.

3. In the setting of $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned} \mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\ &= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\ &\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}. \end{aligned}$$

Here, we have used $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□